National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Appendiceal Neoplasms and Cancers

Version 1.2026 — October 30, 2025

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

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NCCN Guidelines Version 1.2026
Appendiceal Neoplasms and Cancers

NCCN Appendiceal Neoplasms and Cancers Panel Members

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NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

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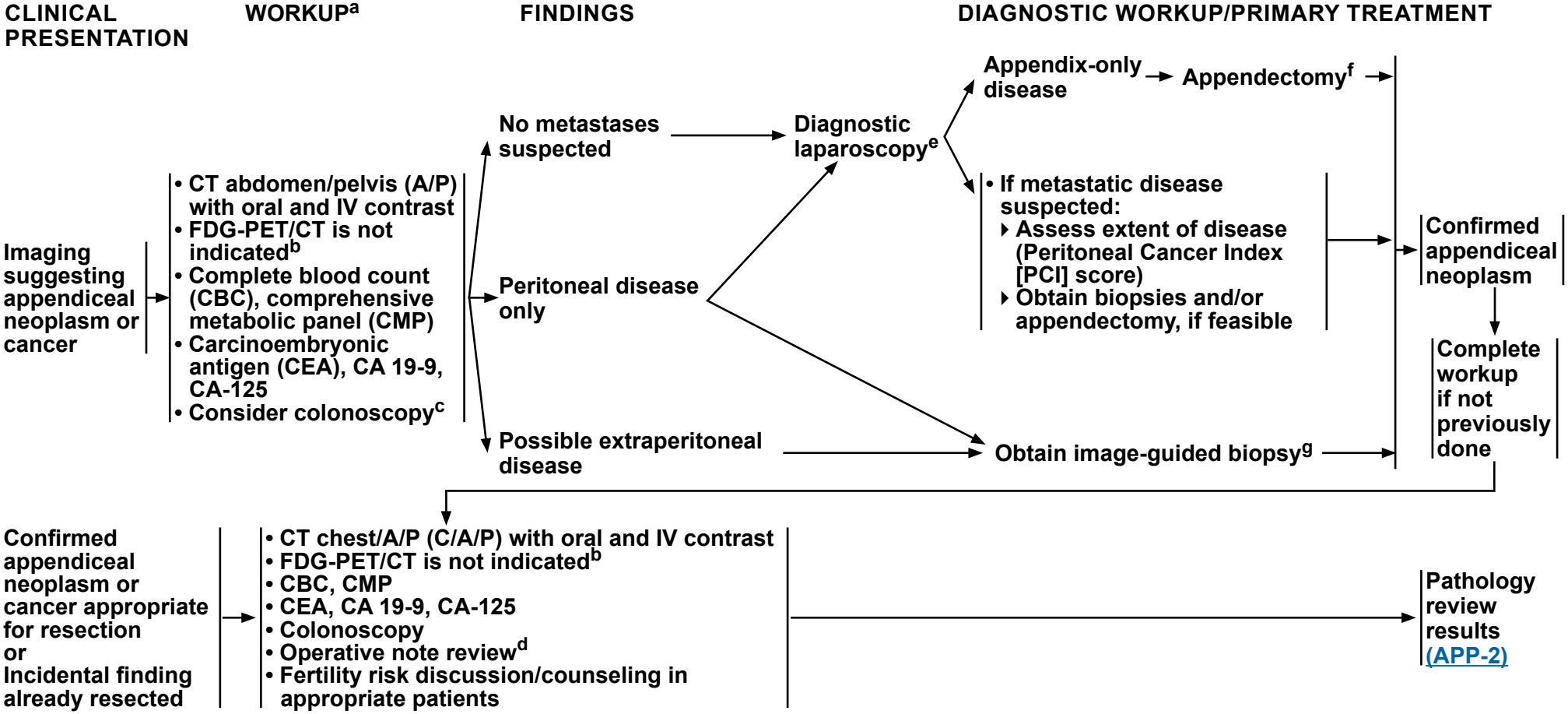

[a] FDA added a Black Box warning to the capecitabine label, recommending to test patients for genetic variants of *DPYD* prior to initiating capecitabine unless immediate treatment is necessary. For information on *DPYD* testing and fluoropyrimidine-associated toxicity, see the NCCN Guidelines for Colon Cancer.

[b] Use of FDG-PET/CT is only considered in rare circumstances where CT findings are indeterminate or when CT with contrast is not possible. See Principles of Imaging (APP-A).

[c] Colonoscopy is needed for a confirmed appendiceal neoplasm to rule out other primary lesions. This could be done concurrently with the other diagnostic studies, but should not delay treatment.

[d] Thorough review of operative notes is encouraged upon confirmation of appendiceal neoplasm in order to rule out more aggressive disease (eg, perforation, ascites present, signs of mucin).

[e] Diagnostic laparoscopy should be performed by a surgeon with experience in this disease process.

[f] See Principles of Surgery and CRS/IPCT (APP-C) for circumstances where a right hemicolectomy may be recommended.

[g] For peritoneal-only disease, image-guided biopsy is preferred for lesions that are safely accessible percutaneously when a non-surgical approach is preferred to confirm the diagnosis (eg, due to comorbidities or extraperitoneal disease). If inaccessible, surgical biopsy may be needed for diagnostic confirmation.

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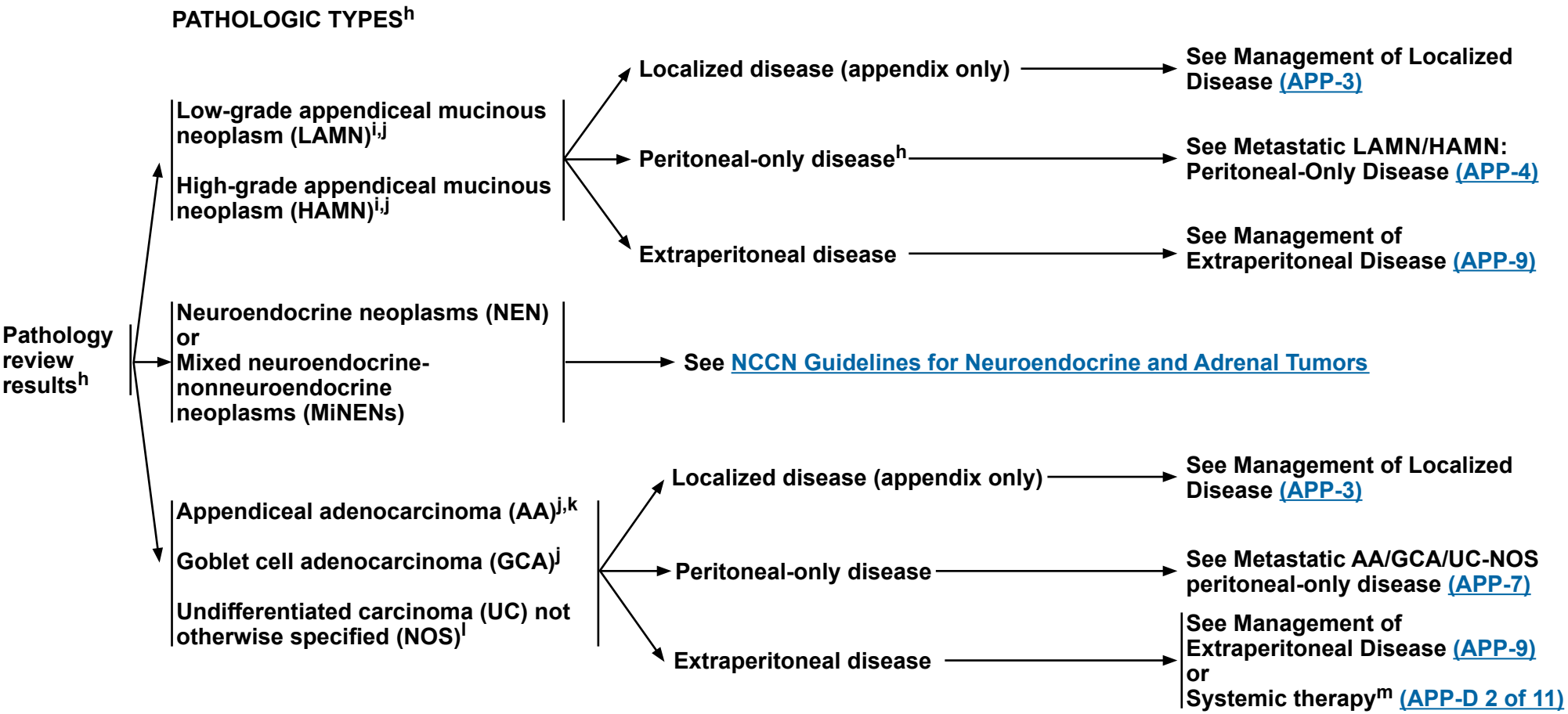

[h] Primary and metastatic disease are evaluated separately. See Principles of Pathology and Molecular Review (APP-B).

[i] Lesions with extra-appendiceal acellular mucin are considered low risk for dissemination. Lesions with extra-appendiceal mucin-containing neoplastic epithelium are considered high risk for recurrence or peritoneal dissemination and the peritoneal-only disease pathway would be followed.

[j] Lesions containing ≤50% signet-ring cells are classified as mucinous adenocarcinomas with signet-ring cells. Tumors containing >50% signet-ring cells are classified as signet-ring cell carcinomas. Signet-ring cell carcinomas are associated with a high level of metastasis and worse prognosis.

[k] Lesions containing ≤50% of mucin are classified as non-mucinous adenocarcinomas. Tumors containing >50% mucin are considered mucinous adenocarcinomas.

[l] UCs only show epithelial marker expression if there is no histologic, immunohistochemical, or molecular indicators of the tissue of origin. Nagtegaal ID, et al. Histopathology 2020;76:182-188.

[m] Biomarker testing is recommended, although there may be a higher rate of assay failure if specimen has low cellularity. See Principles of Pathology and Molecular Review (APP-B).

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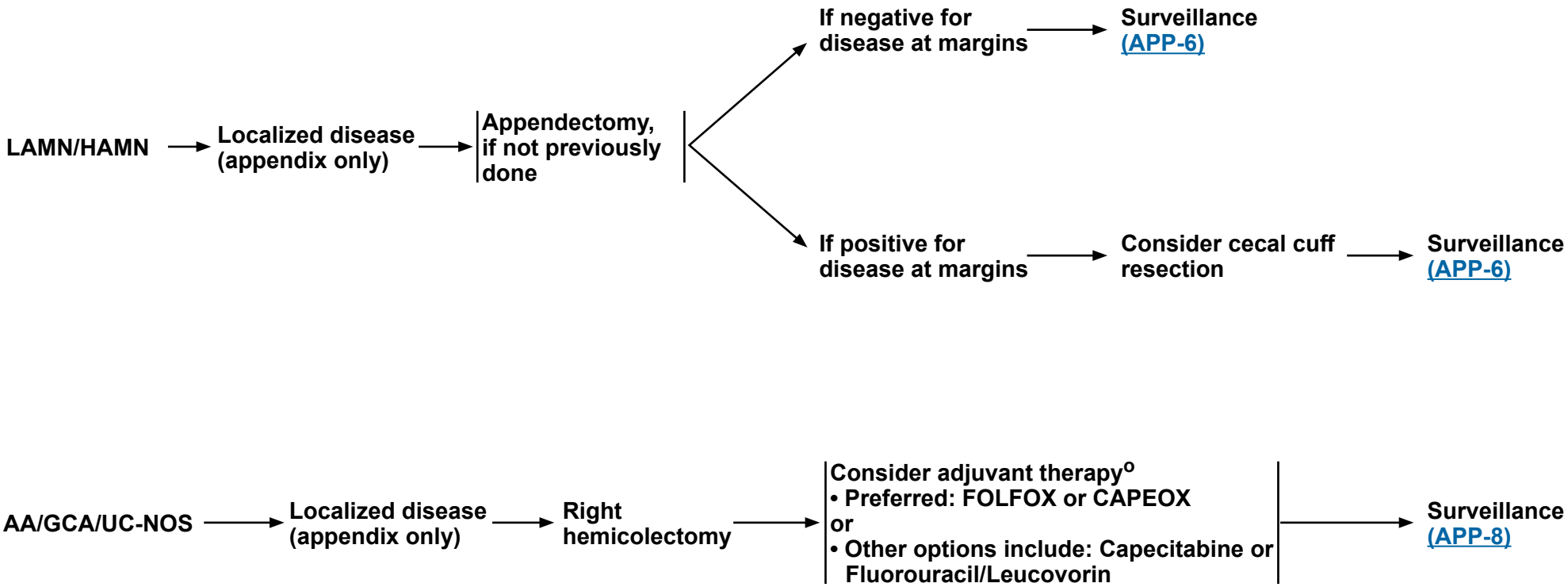
MANAGEMENT OF LOCALIZED DISEASE[n]



[n] Multidisciplinary team evaluation, including formal surgical evaluation.
[o] Strongly consider adjuvant chemotherapy for patients with nodal disease or perforation noted at the time of surgery. May be considered in other instances depending on high-risk features (extrapolating from colon cancer, these features may be: poorly differentiated/undifferentiated histology; lymphatic/vascular invasion; bowel obstruction; <12 lymph nodes examined; perineural invasion [PNI]; localized perforation; close, indeterminate, positive margins; or high-tier tumor budding). Circulating tumor (ctDNA) is prognostic, but not predictive.

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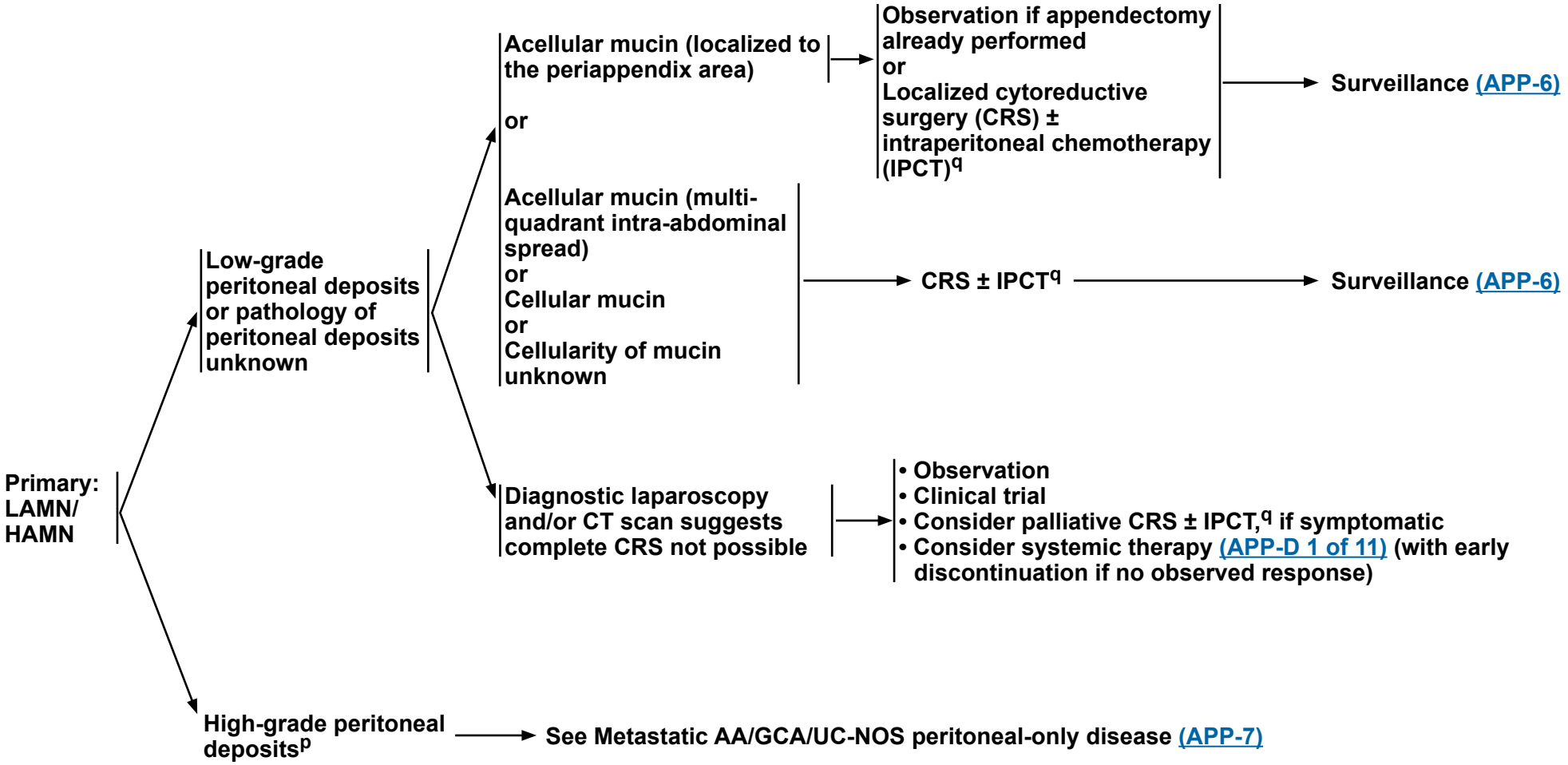
METASTATIC LAMN/HAMN: PERITONEAL-ONLY DISEASE[n]



[n] Multidisciplinary team evaluation, including formal surgical evaluation.
[p] High-grade peritoneal deposits behave clinically like high-risk disease and should follow those treatment pathways, even if the primary is low grade.
[q] Principles of Surgery and CRS/IPCT (APP-C).

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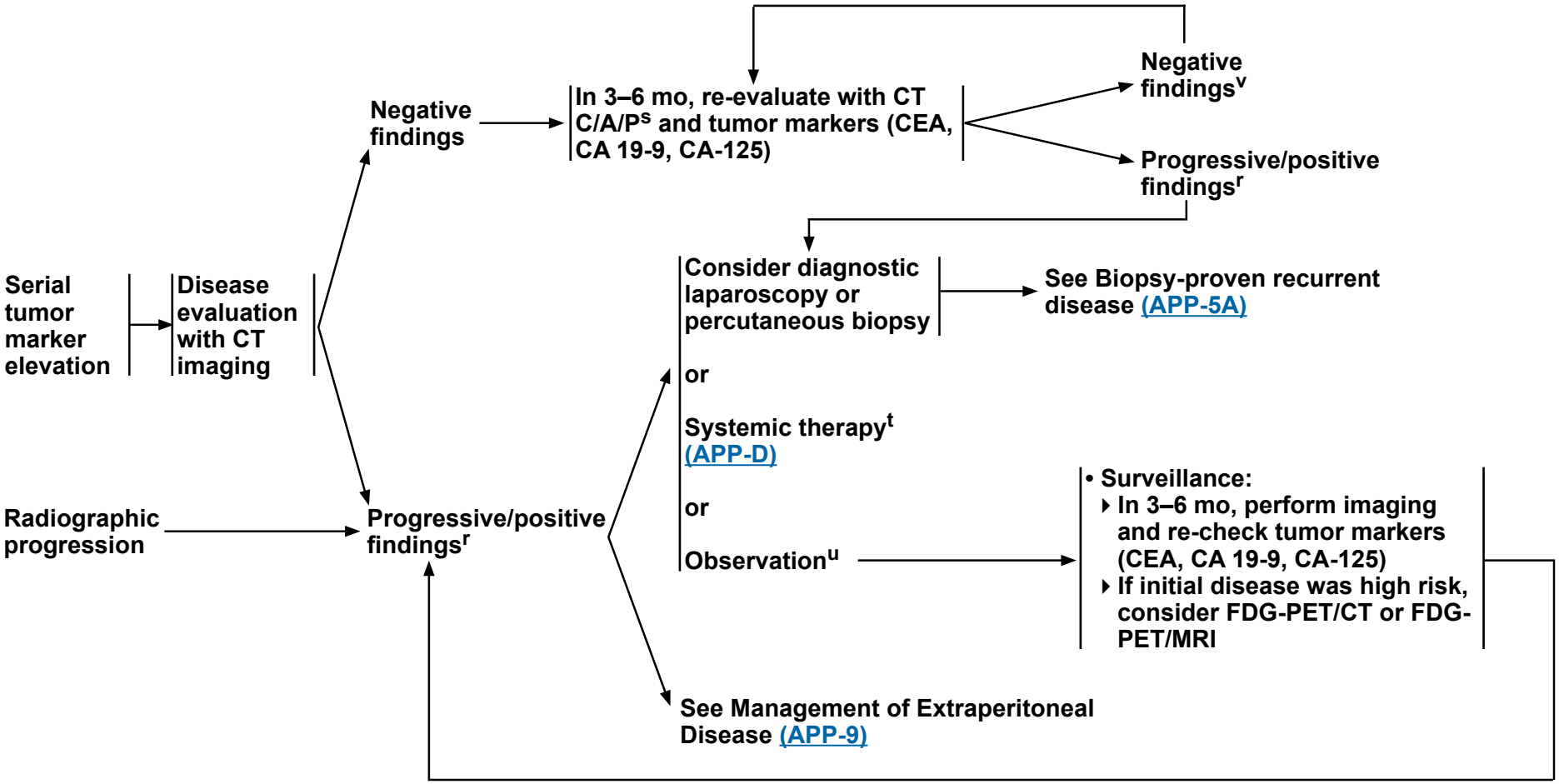
WORKUP FOR RECURRENCE[n]



[n] Multidisciplinary team evaluation, including formal surgical evaluation.
[r] ctDNA assays are not recommended in this setting. These assays have a lower sensitivity for peritoneal disease and false negatives are common.
[s] If CT is normal and disease is considered high risk at initial diagnosis, consider FDG-PET/CT or FDG-PET/MRI.
[t] For progressive disease in a patient who is not a surgical candidate, systemic therapy may be considered without a biopsy. Follow the systemic therapy pathway according to the risk level of histology.
[u] In select cases per clinical judgment. For low-risk disease and indolent changes, observation with continued disease assessment may be considered.
[v] If consistently negative findings on multiple rounds of re-evaluation, interval may be lengthened to every 6–12 months.

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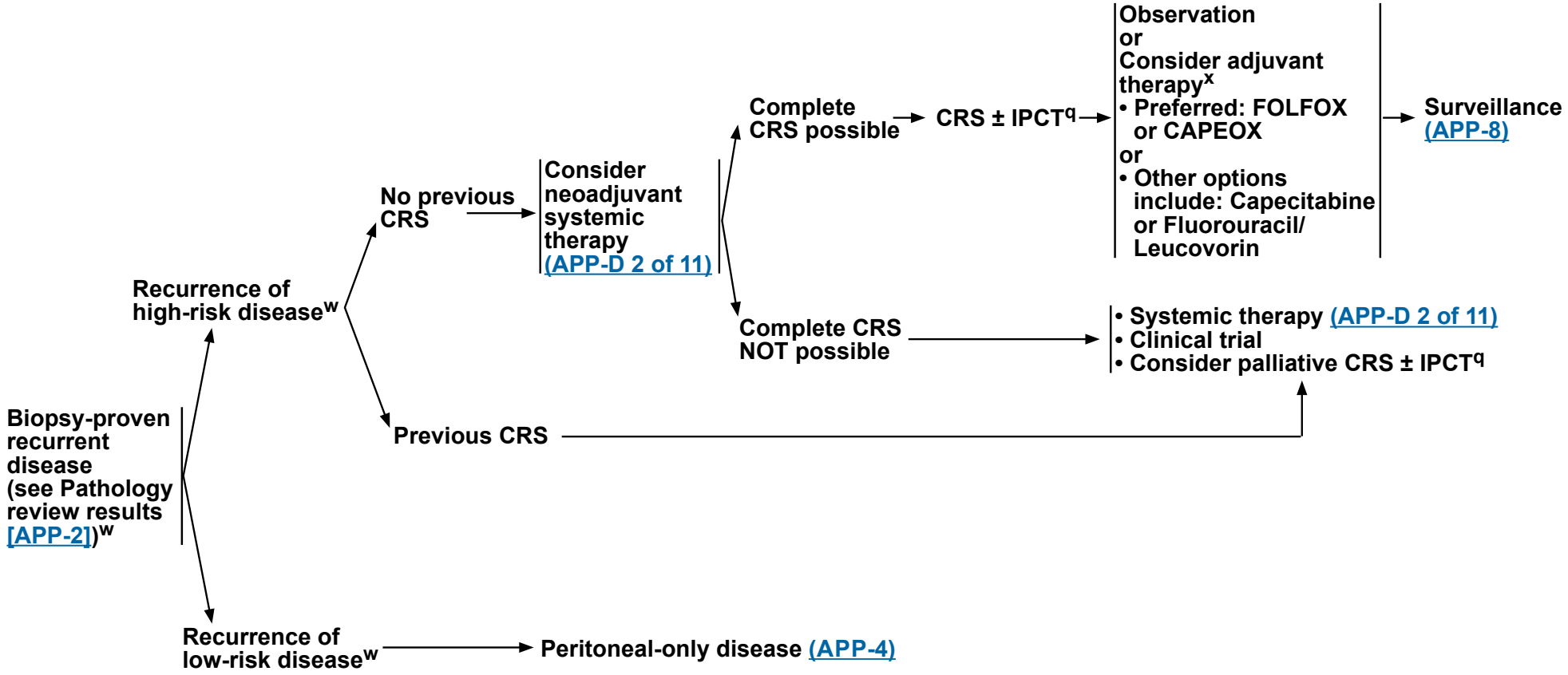
BIOPSY-PROVEN RECURRENCE[n]



[n] Multidisciplinary team evaluation, including formal surgical evaluation.
[q] Principles of Surgery and CRS/IPCT (APP-C).
[w] Risk status should be according to pathology of metastatic disease; subsequent treatment should be based on the highest risk pathology found (thus, a low-risk primary with high-risk metastatic disease should be treated as per high-risk pathways).
[x] This recommendation has been extrapolated from colon cancer; duration of perioperative therapy should be limited to no more than 6 months.

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SURVEILLANCE: LAMN/HAMN AND NO HIGH-GRADE PERITONEAL DEPOSIT

Resected localized disease[y] →

- History and physical examination every 6–12 mo for 2 y, then annually for up to 10 y
- CEA, CA 19-9, CA-125 every 6–12 mo for 2 y, then annually for up to 10 y[z]
- A/P CT or MRI every 6–12 mo from date of surgery for 2 years then every 12–24 mo for up to 10 y
- FDG-PET/CT is not indicated
- Colonoscopy, if no preoperative colonoscopy was done
- Principles of Survivorship (APP-E)

Resected metastatic disease →

- History and physical examination every 6 mo for 2 y, then annually for up to 10 y
- CEA, CA 19-9, CA-125 every 6–12 mo for 2 y, then annually for up to 10 y[z]
- C/A/P CT or MRI every 6 mo from date of surgery for 2 years then every 12 mo for up to 10 y
- FDG-PET/CT is not indicated
- Colonoscopy, if no preoperative colonoscopy was done
- Principles of Survivorship (APP-E)

→ **Rising tumor markers, unexplained abdominal symptoms, documented recurrence, or radiographic progression** → **See Recurrence (APP-5)**

[y] For completely resected LAMN confined to the appendix without high-risk features (eg, extra-appendiceal cellular mucin, presence of signet ring cells), surveillance may not be needed.
[z] It may not be necessary to measure all tumor markers at all surveillance timepoints as guided by prior marker elevations for the specific patient.

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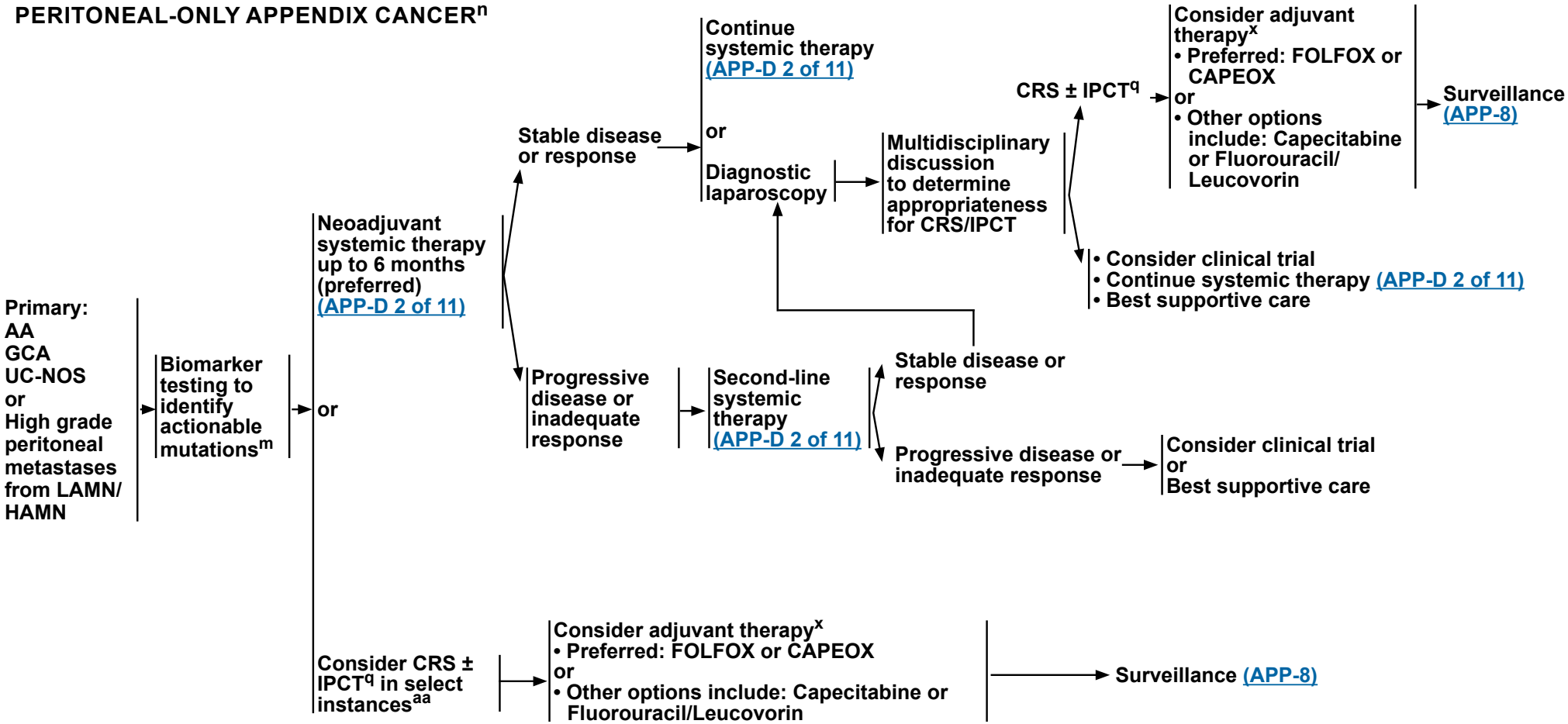
METASTATIC AA/GCA/UC-NOS PERITONEAL-ONLY APPENDIX CANCER[n]



[m] Biomarker testing is recommended, although there may be a higher rate of assay failure if specimen has low cellularity. See Principles of Pathology and Molecular Review (APP-B).
[n] Multidisciplinary team evaluation, including formal surgical evaluation.
[q] Principles of Surgery and CRS/IPCT (APP-C).
[x] This recommendation has been extrapolated from colon cancer; duration of perioperative therapy should be limited to no more than 6 months.
[aa] Select instances including low disease burden and/or well-differentiated adenocarcinoma.

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SURVEILLANCE: AA/GCA/UC-NOS

Resected localized disease →
- History and physical examination every 3–6 mo for 2 y, then every 6 mo for a total of 5 y
- CEA, CA 19-9, CA-125 every 3–6 mo for 2 y, then 6–12 mo for a total of 5 y
- C/A/P CT every 6–12 mo from date of surgery for a total of 5 y
- FDG-PET/CT is not indicated
- Colonoscopy, if no preoperative colonoscopy was done
- Principles of Survivorship (APP-E)

Resected metastatic disease, including resected LAMN/HAMN primary with high-grade metastatic disease →
- History and physical examination every 3–6 mo for 2 y, then every 6 mo for a total of 5 y
- CEA, CA 19-9, CA-125 every 3–6 mo for 2 y, then every 6 mo for a total of 5 y
- C/A/P CT every 3–6 mo for 2 years then every 6–12 mo for a total of 5 y
- FDG-PET/CT is not indicated
- Colonoscopy, if no preoperative colonoscopy was done
- Principles of Survivorship (APP-E)

→ **Rising tumor markers, unexplained abdominal symptoms, or documented recurrence** → **See Recurrence (APP-5)**

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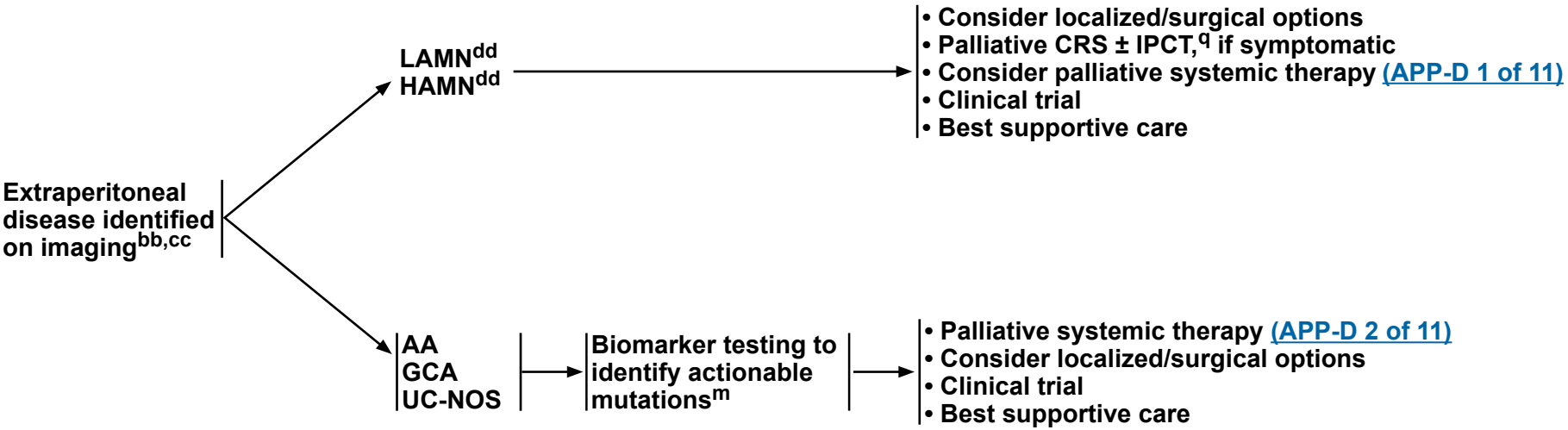
MANAGEMENT OF EXTRAPERITONEAL DISEASE[n]



[m] Biomarker testing is recommended, although there may be a higher rate of assay failure if specimen has low cellularity. See Principles of Pathology and Molecular Review (APP-B).
[n] Multidisciplinary team evaluation, including formal surgical evaluation.
[q] Principles of Surgery and CRS/IPCT (APP-C).
[bb] If CT is not feasible, consider MRI of abdomen to differentiate between liver and extrahepatic disease.
[cc] Specify whether the metastases is within the liver parenchyma (extraperitoneal metastasis) or scalloping along the liver edge (still intraperitoneal metastasis).
[dd] It would be extremely unusual for low-risk disease to have extraperitoneal spread; therefore, pathologic confirmation is recommended to rule out high-risk disease.

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PRINCIPLES OF IMAGING[1-5]

- It is important to note that imaging may be negative in the setting of peritoneal metastases, or that it may underestimate the peritoneal disease volume. Direct visualization (laparoscopic evaluation when feasible) should be performed for confirmation in the staging setting and when recurrence is suspected clinically and imaging findings are negative.
- The mainstay of staging imaging should consist of a CT of the C/A/P OR non-contrast CT of the chest with MRI of the A/P.
  - Evaluate local extent of tumor, presence of associated inflammation and/or perforation, and infiltration of adjacent structures and regional metastases.
  - Evaluate for distant metastases (ie, lymph nodes, lung, liver, peritoneum, other organs)
  - CT exam should be performed with oral and IV iodinated contrast, unless contraindicated.
  - If IV iodinated contrast is contraindicated due to severe allergy, then CT of the chest without contrast and MRI of the A/P with IV gadolinium-based contrast agent (GBCA) can be obtained instead.
  - In patients with chronic renal failure (estimated glomerular filtration rate [eGFR] <30 mL/min) who are not on dialysis, IV contrast material is also contraindicated, and IV GBCA such as gadofosveset trisodium, gadoxetate disodium, gadobenate dimeglumine, or gadoteridol may be used.
  - If both IV iodinated contrast and IV GBCA are contraindicated, CT of the chest without contrast and MRI of the A/P without contrast should be considered. PET/CT may also be considered as an alternate.
  - Consider MRI of the liver in cases where better resolution of indeterminate liver lesions is needed and/or MRI of select cases of hepatic metastases when planning metastasectomy or liver-directed therapies. In these cases, utilization of hepatocyte-specific GBCA is preferred for precise localization and quantification of metastases.

- FDG-PET/CT
  - FDG-PET/CT is usually not indicated. In select cases (ie, neuroendocrine neoplasm), DOTATATE PET/CT should be considered.
    - FDG-PET does not supplant a contrast-enhanced diagnostic CT or MRI and can be used to evaluate an equivocal finding on a contrast-enhanced CT or MRI, but should not be the preferred imaging study.
  - FDG-PET may under-appreciate disease burden, especially for mucinous tumors.

- Monitoring
  - For treatment response monitoring, typically a CT of the A/P (in cases of resected localized disease), or CT of the C/A/P (in cases of resected metastatic disease) with oral and IV contrast, or non-contrast CT of the chest with MRI of the A/P should be used.
    - For patients who have had liver-directed therapy (ablation-based, ie, radiofrequency ablation [RFA], microwave ablation [MWA]; radiation-based, ie, radioembolization, stereotactic body radiation therapy [SBRT]), MRI of the liver is preferred over CT for the assessment of treatment response.
    - In select cases FDG-PET may be indicated to assess residual tumor viability if curative liver resection is contemplated, but should be done in partnership with a contrasted CT or MRI.
    - For mucinous tumors, MRI of the A/P may be preferred over CT as mucinous implants may be easier to distinguish from adjacent structures due to their T2 signal.

References

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PRINCIPLES OF IMAGING
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PRINCIPLES OF PATHOLOGY AND MOLECULAR REVIEW

Appendiceal Cancer Appropriate for Resection

- **Primary appendiceal neoplasms are rare and most often found incidentally during an appendectomy following clinical presentation of acute appendicitis.**
- **Individual management and treatment of primary appendiceal cancers are dependent on classification, grading, and staging.**
- **According to the most recent WHO classification of tumors, appendiceal epithelial neoplasms are categorized as follows[1]:**
  - **Serrated lesions and polyps**
  - **Mucinous neoplasm**
  - **Adenocarcinoma**
  - **GCA**
  - **Neuroendocrine neoplasms**

Background: Appendiceal Mucinous Neoplasms

- **There is much ongoing debate over terminology and classification of mucinous neoplasms due to the fact that even low-grade neoplasms can result in pseudomyxomatous peritonei and lack of consensus criteria for invasion into the appendix.**
- **Current terminology includes low-grade appendiceal mucinous neoplasm (LAMN), high-grade appendiceal mucinous neoplasm (HAMN), low-grade mucinous carcinoma peritonei, high-grade mucinous carcinoma peritonei, and high-grade mucinous carcinoma peritonei with signet ring cells.**
- **Older terminology that is no longer recommended includes: mucinous tumor of uncertain malignant potential, mucinous cystadenoma of the appendix, borderline tumor of the appendix, mucinous cystadenocarcinoma of the appendix, disseminated peritoneal adenomucinosis, and peritoneal mucinous carcinomatosis with or without signet ring cells.[1,2,3]**
- **However, some pathologists prefer to use the term "mucinous cystadenoma" to refer to entirely submitted mucinous tumors with no mural or extra-appendiceal mucin. If mural changes are present, such as disruption or atrophy of the muscularis mucosa, these lesions are best classified as LAMN.[4,5]**
  - **Low-grade appendiceal mucinous neoplasm (LAMN):**
    - ◊ **LAMNs are characterized by tall mucinous epithelial cells in villous or flat arrangement, which invade with a distinct pushing border in contrast to infiltrative glands associated with desmoplasia.**
    - ◊ **LAMNs are most often associated with atrophy of the appendiceal wall, loss of lymphoid tissue, and effacement of the muscularis mucosa, which can be helpful in differentiating other entities on the histologic differential diagnosis such as ruptured diverticula.[4,5]**
    - ◊ **Mucin pools are common and can extend into the serosa and beyond.**
    - ◊ **According to the 9th version of the AJCC, LAMNs confined to the muscularis propria, including acellular mucin or mucinous epithelium, are staged as carcinoma in situ (pTis). pT1 and pT2 are not applicable in LAMN.**
    - ◊ **Acellular mucin or mucinous epithelium that extends into the subserosa or serosa is classified as pT3 or pT4a, respectively.[6]**
    - ◊ **LAMNs are almost always microsatellite stable (MSS), *BRAF* V600E negative, and frequently harbor *KRAS* mutations.[2]**

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PRINCIPLES OF PATHOLOGY AND MOLECULAR REVIEW

- **High-grade appendiceal mucinous neoplasm (HAMN):**
  - **HAMNs retain some similar findings as LAMN but show progression to a more aggressive lesion with the acquisition of high-grade cytologic features and complex architecture.**
  - **As such, HAMNs are staged as mucinous adenocarcinomas, although HAMNs are rare and robust data are limited.[2]**
- **Mucinous carcinoma peritonei:**
  - **Extra-appendiceal mucin is characterized as acellular or with neoplastic cells that can be low grade, high grade, or high grade with signet ring cells.**
  - **The presence or absence of neoplastic cells and the degree of cytologic atypia have prognostic implications in that acellular mucin confined to the right lower quadrant has a low risk of progression whereas mucin with high-grade neoplastic epithelium with signet ring cells is associated with a much worse prognosis.[2,7]**
- **Appendiceal adenocarcinoma (AA):**
  - **Outside of LAMN and pseudomyxoma peritonei, other primary AAs are rare and can arise from several distinct precursor lesions such as mucinous neoplasms, intestinal-type adenomas, and GCAs.**
  - **These adenocarcinomas often demonstrate abundant extracellular mucin but are distinguished from mucinous neoplasms by a few key differences.**
  - **These tumors are highly cellular, have high-grade nuclear features, and show an infiltrative growth pattern with desmoplastic response.[8,9]**
  - **Primary mucinous adenocarcinomas are rarely microsatellite unstable (microsatellite instability-high [MSI-H]). However, non-mucinous subtypes have been shown to be MSI-H.[10]**
  - **Mucinous subtypes are associated with better outcomes than non-mucinous subtypes. However, higher histologic grade in mucinous subtypes is associated with lower overall 5-year survival rates.[11]**
- **Appendiceal goblet cell adenocarcinoma (GCA):**
  - **These tumors have been historically classified as NETs but are now recognized as a distinct clinicopathologic entity with biologic potential more similar to adenocarcinomas than NETs.[9]**
  - **Older terminology includes goblet cell carcinoid, goblet cell type adenocarcinoid, mucinous carcinoid, and crypt cell carcinoma.**
  - **Microscopically, GCAs are characterized by deep mucosal and submucosal circumferential infiltration without associated surface epithelial dysplasia.**
  - **Extension into the muscularis propria and serosa is commonly observed.**
  - **GCAs are defined by the presence of a low-grade component comprised of small nests, clusters, and cords of goblet cells. Paneth cells and neuroendocrine cells may be present but are not required for a diagnosis of GCA.**
  - **The presence of infiltrating single or small clusters of mucinous or non-mucinous cells, complex cribriform glands, or solid sheets of cells or signet ring cells are evidence of a high-grade component.**

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PRINCIPLES OF PATHOLOGY AND MOLECULAR REVIEW

- **Appendiceal goblet cell adenocarcinoma (GCA) (continued):**
  - ◊ **Recognizing low- and high-grade patterns is critical, as grading GCA depends on the percentages of each component.**
    - **– Grade 1 and 2 tumors demonstrate >75% and 50%–75% of tubular or clustered growth, respectively.**
    - **– Grade 3 tumors reveal <50% low-grade pattern and >50% of any combination of the high-grade pattern.[12,13]**
  - ◊ **As GCAs are often deeply invasive, patients with low-grade histology may present with evidence of metastasis.**
  - ◊ **Interestingly, GCAs have different molecular profiles from other AAs and neuroendocrine neoplasms in that lower rates of *KRAS* and *APC* mutations are consistently reported. However, microsatellite instability (MSI) and high tumor mutational burden (TMB-H) are also rare in these tumors.[14,15]**
- **Appendiceal neuroendocrine neoplasms:**
  - ◊ **Neuroendocrine neoplasms arising in the appendix include well-differentiated NETs, poorly differentiated neuroendocrine carcinomas (NECs), and MiNENs.[16]**
  - ◊ **Please see NCCN Guidelines for Neuroendocrine and Adrenal Tumors.**

Biomarker Testing

- **Biomarker testing, including:**
  - ***RAS* and *BRAF* V600E mutations; HER2 status; mismatch repair (MMR) or MSI status (if not previously done)**
  - **Testing should be conducted as part of multigene panel testing (MGPT), which would identify rare and actionable mutations and fusions such as *POLE/POLD1, RET,* and *NTRK 1/2/3*.**
  - **For more information on biomarker testing, see the Principles of Pathologic and Molecular Review in the NCCN Guidelines for Colon Cancer.**

References

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PRINCIPLES OF SURGERY AND CRS/IPCT

- **Localized Disease:**
  - **Appendiceal tumors are heterogenous groups of tumors. Knowledge about the different pathologic types is key to appropriate surgical management.**
  - **Appendiceal tumors are rare but they are more common among patients with complicated appendicitis.**
  - **Histologies at lower risk of hematogenous spread, including LAMN and HAMN:**
    - **Negative margin appendectomy is largely considered to be sufficient. Ileocecectomy may be considered when appendectomy or extended appendectomy is not feasible.**
    - **Management of HAMN is controversial with some advocating for right colectomy.**
    - **Presence of perforation is a risk factor for peritoneal recurrence and warrants closer surveillance.**
  - **Histologies at higher risk of hematogenous spread, including adenocarcinoma:**
    - **Right hemicolectomy with adequate lymphadenectomy is recommended.**
  - **There is no clear role for routine second look laparoscopy in the absence of other suggestive risk factors for recurrent disease.**

- **Peritoneal Disease:**
  - **In cases of incidental peritoneal mucinous or non-mucinous disease during appendectomy or surgical exploration, the focus should be on obtaining a diagnosis rather than extensive resection. Diagnosis can be confirmed by appendectomy and/or biopsy of the most solid-appearing peritoneal implants.**
  - **Preoperative imaging may underestimate the extent of disease in the peritoneal cavity.**
  - **Staging laparoscopic exploration may be helpful in estimating the extent of peritoneal disease (PCI) and extent of surgery that will be required.**
  - **Patients deemed possible surgical candidates should be evaluated and treated by an experienced peritoneal surface malignancy surgeon for candidacy for CRS/IPCT. Expertise in peritoneal resections, multivisceral resections, and organ preservation techniques is key.**
    - **Candidacy for CRS/IPCT is determined through a multidisciplinary evaluation of the patient's overall health, disease biology and extent, and tumor resectability.**
  - **The extent of CRS should be individualized with the focus on optimal cytoreduction with organ preservation.**
  - **For surgical candidates, optimal cytoreduction (not debulking) with completeness of cytoreduction (CC) score of 0 or 1 is recommended.**
  - **Mitomycin C is the most commonly used agent for IPCT. Oxaliplatin is another option, but is less preferred, especially in patients who have received or will be receiving IV oxaliplatin.[1,2]**
  - **Repeat CRS/IPCT may be considered for patients with a history of previous CRS, particularly those with low-grade peritoneal tumors and/or limited peritoneal spread. For high-grade disease, the time since prior surgery should be taken into consideration.**
  - **In cases where disease is incompletely cytoreduced or resected or CRS is aborted, consider clinical trial, systemic therapy, or best supportive care.**

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PRINCIPLES OF SURGERY AND CRS/IPCT

- **No Extraperitoneal Metastasis:**
  - **LAMN/HAMN Primary:**
    - ◊ **The majority of mucinous appendiceal (non-adenocarcinoma) neoplasms are less likely to respond to systemic therapy.**
    - ◊ **CRS/IPCT is recommended for low-grade (and select high-grade) mucinous tumors.**
    - ◊ **Optimal cytoreduction should be pursued when possible.**
    - ◊ **Palliative CRS/IPCT may be considered for unresectable disease in patients who are otherwise surgical candidates.**
  - **AA/GCA/UC-NOS:**
    - ◊ **Surgical candidates are suggested to receive perioperative systemic therapy for up to 6 months.**
    - ◊ **Regular multidisciplinary review is recommended to assess surgical candidacy and optimal timing of surgery, recognizing that patients may have poor response to systemic therapy but still have disease that is amenable to CRS.**
    - ◊ **Additional systemic therapy may be considered for patients who are not resectable at initial diagnosis with the possibility of converting to resectable disease.**
    - ◊ **The PCI score and CC score should be reported for cytoreductive surgery.**
    - ◊ **Surgery is discouraged for high PCI.**
    - ◊ **If a patient is not a candidate for surgery, treatment should follow metastatic colon cancer guidelines.**
- **Extraperitoneal Disease:**
  - **Liver surface metastasis (scalloping) is not considered extraperitoneal disease.**
  - **Management should follow systemic therapy guidelines with palliative measures used as needed for refractory symptoms.**
  - **Palliative debulking surgery can be considered for disease not responding or poorly responding to systemic therapy, but may be of limited benefit.**

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PRINCIPLES OF PALLIATIVE SYSTEMIC THERAPY FOR LAMN OR HAMN[*,a,b,c]

INITIAL THERAPY
• Capecitabine ± Bevacizumab • Fluorouracil/Leucovorin ± Bevacizumab • FOLFIRI (Leucovorin/Fluorouracil/Irinotecan)[d] ± Bevacizumab • FOLFOX (Leucovorin/Fluorouracil/Oxaliplatin)[e] ± Bevacizumab

For AA/GCA/UC-NOS, see APP-D 2 of 11
Footnotes APP-D 3 of 11

[*] Prolonged chemotherapy exposure is not recommended for patients who are not demonstrating a clinical response.

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PRINCIPLES OF PALLIATIVE SYSTEMIC THERAPY FOR AA/GCA/UC-NOS[a,b,c,f]

INITIAL THERAPY

Intensive Therapy Recommended

- CAPEOX (Capecitabine/Oxaliplatin)[e] ± Bevacizumab
- FOLFIRI[d] ± Bevacizumab
- FOLFIRINOX (Leucovorin/Fluorouracil/Irinotecan/Oxaliplatin)[e,g] ± Bevacizumab
- FOLFOX[e] ± Bevacizumab

- *BRAF* V600E mutation positive:
  - FOLFOX[e]/Encorafenib + (Cetuximab or Panitumumab)

Intensive Therapy NOT Recommended

- Capecitabine ± Bevacizumab
- Fluorouracil/Leucovorin ± Bevacizumab

FOR ANY LINE OF THERAPY (if dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])[h,i,j]

- Cemiplimab-rwlc
- Dostarlimab-gxly
- Ipilimumab + Nivolumab[k]
- Nivolumab[k]
- Pembrolizumab[l]
- Penpulimab-kcqx
- Retifanlimab-dlwr
- Tislelizumab-jsgr
- Toripalimab-tpzi

SECOND-LINE AND SUBSEQUENT THERAPY (if not previously given)

- (FOLFIRI or Irinotecan)[d] ± Bevacizumab
- FOLFIRINOX[e,g] ± Bevacizumab
- (CAPEOX or FOLFOX)[e] ± Bevacizumab
- IROX (Irinotecan[d]/Oxaliplatin[e]) ± Bevacizumab

- For disease that has progressed through all available regimens:
  - Fruquintinib
  - Regorafenib
  - Trifluridine and Tipiracil ± Bevacizumab (Bevacizumab combo preferred)

- Best supportive care (NCCN Guidelines for Palliative Care)

Biomarker-directed therapy

- *BRAF* V600E mutation positive
  - Encorafenib + (Cetuximab or Panitumumab)

- *KRAS* G12C mutation positive
  - (Adagrasib or Sotorasib)[m] + (Cetuximab or Panitumumab)

- HER2-positive and *RAS* and *BRAF* WT
  - (Pertuzumab or Tucatinib)[n] + Trastuzumab[n]

- HER2-positive (IHC 3+)
  - Fam-trastuzumab deruxtecan-nxki[o]

- *NTRK 1/2/3* gene fusion-positive
  - Entrectinib
  - Larotrectinib
  - Repotrectinib[p]

- *RET* gene fusion-positive
  - Selpercatinib

- If *KRAS/NRAS/BRAF* WT[q]:
  - CAPEOX[e] + (Cetuximab or Panitumumab)
  - (Cetuximab or Panitumumab)[r] ± Irinotecan[d]
  - FOLFIRI[d] + (Cetuximab or Panitumumab)[r]
  - FOLFOX[e] + (Cetuximab or Panitumumab)

Footnotes APP-D 3 of 11

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PRINCIPLES OF PALLIATIVE SYSTEMIC THERAPY
FOOTNOTES

[a] For systemic therapy references, see Regimens and Dosing (APP-D 4 of 11).
[b] For infection risk, monitoring, and prophylaxis recommendations for targeted therapies, see INF-A in the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections.
[c] An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.
[d] Irinotecan should be used with caution in patients with Gilbert syndrome or elevated serum bilirubin. There is a commercially available test for *UGT1A1*. Guidelines for use in clinical practice have not been established.
[e] Discontinuation of oxaliplatin should be strongly considered after 3 to 4 months of therapy (or sooner for unacceptable neurotoxicity) while maintaining other agents until time of progression. Oxaliplatin may be reintroduced if it was discontinued for neurotoxicity rather than for disease progression.
[f] There is no clear evidence for anti-EGFR therapy, even among *RAS/RAF* WT cancers.
[g] FOLFIRINOX is recommended instead of FOLFOXIRI because FOLFOXIRI uses a high dose of fluorouracil (3,200 mg/m² over 48 hours). Patients in the United States have been shown to have greater toxicity with fluorouracil. The dose of fluorouracil (2,400 mg/m² over 46 hours) is a starting dose consistent with the dose recommended in FOLFOX or FOLFIRI and should be strongly considered for U.S. patients.
[h] NCCN Guidelines for Management of Immunotherapy-Related Toxicities.
[i] If disease response, consider discontinuing checkpoint inhibitor after 2 years of treatment.
[j] If no previous treatment with a checkpoint inhibitor. Nivolumab + ipilimumab may be considered as subsequent therapy if checkpoint inhibitor monotherapy was previously received.
[k] Nivolumab and hyaluronidase-nvhy is not approved for concurrent use with IV ipilimumab; however, for nivolumab monotherapy, nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for IV nivolumab. Nivolumab and hyaluronidase-nvhy has different dosing and administration instructions compared to IV nivolumab.
[l] Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.
[m] If patient is unable to tolerate EGFR inhibitor due to toxicity, single-agent adagrasib or sotorasib can be considered.
[n] If no previous treatment with HER2 inhibitor.
[o] Some activity was seen after a previous HER2-targeted regimen. May not be indicated in patients with underlying lung issues due to lung toxicity (3.5% report of drug-related deaths from interstitial lung disease on the DESTINY-CRC01 trial).
[p] On the TRIDENT-1 trial, repotrectinib showed activity in both NTRK TKI-naïve and NTRK TKI-pretreated patients.
[q] Patients with *BRAF* mutations other than V600E may be considered for anti-EGFR therapy.
[r] Cetuximab or panitumumab are recommended in combination with irinotecan-based therapy or as single-agent therapy for patients who cannot tolerate irinotecan.

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PRINCIPLES OF PALLIATIVE SYSTEMIC THERAPY – REGIMENS AND DOSING[c]

Modified FOLFOX6[1,2,3]
Oxaliplatin 85 mg/m² IV Day 1[s]
Leucovorin 400 mg/m² IV Day 1[t]
Fluorouracil 400 mg/m² IV bolus Day 1, followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) IV continuous infusion
Repeat every 2 weeks

Modified FOLFOX7[4]
Oxaliplatin 85 mg/m² IV Day 1[s]
Leucovorin 400 mg/m² IV Day 1[t]
Fluorouracil 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) IV continuous infusion
Repeat every 2 weeks

FOLFOX + Bevacizumab[5,u]
Bevacizumab 5 mg/kg IV Day 1
Repeat every 2 weeks

FOLFOX + Panitumumab[6] (*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes, Day 1
Repeat every 2 weeks

FOLFOX + Cetuximab[7] (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² IV over 2 hours first infusion, followed by 250 mg/m² IV over 60 minutes weekly or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks (preferred for every 2 weeks)

CAPEOX[8]
Oxaliplatin 130 mg/m² IV Day 1[s]
Capecitabine 1000[v] mg/m² twice daily PO for 14 days
Repeat every 3 weeks

CAPEOX + Bevacizumab[8,u]
Oxaliplatin 130 mg/m² IV Day 1[s]
Capecitabine 1000[v] mg/m² PO twice daily for 14 days
Bevacizumab 7.5 mg/kg IV Day 1
Repeat every 3 weeks

CAPEOX + Cetuximab[9-11] (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² IV over 2 hours first infusion, followed by 250 mg/m² IV over 60 minutes weekly or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks (preferred for every 2 weeks)

CAPEOX + Panitumumab[9-11] (*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes, Day 1
Repeat every 2 weeks

FOLFIRI[12,13]
Irinotecan 180 mg/m² IV over 30–90 minutes, Day 1
Leucovorin[t] 400 mg/m² IV infusion to match duration of Irinotecan infusion, Day 1
Fluorouracil 400 mg/m² IV bolus Day 1, followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) continuous infusion
Repeat every 2 weeks

FOLFIRI + Bevacizumab[14,u]
Bevacizumab 5 mg/kg IV Day 1
Repeat every 2 weeks

References

[c] An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.
[s] Oxaliplatin may be given either over 2 hours, or may be infused over a shorter time at a rate of 1 mg/m²/min. Leucovorin infusion should match infusion time of oxaliplatin. Cercek A, Park V, Yaeger R, et al. Faster FOLFOX: oxaliplatin can be safely infused at a rate of 1 mg/m²/min. J Oncol Pract 2016;12:e548-553.
[t] Leucovorin 400 mg/m² is the equivalent of levoleucovorin 200 mg/m².
[u] Bevacizumab may be safely given at a rate of 0.5 mg/kg/min (5 mg/kg over 10 minutes and 7.5 mg/kg over 15 minutes).
[v] The majority of safety and efficacy data for this regimen have been developed in Europe, where a capecitabine starting dose of 1000 mg/m² twice daily for 14 days, repeated every 21 days, is standard. Evidence suggests that North American patients may experience greater toxicity with capecitabine (as well as with other fluoropyrimidines) than European patients, and may require a lower dose of capecitabine.

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PRINCIPLES OF PALLIATIVE SYSTEMIC THERAPY – REGIMENS AND DOSING[c]

FOLFIRI + Cetuximab (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² IV over 2 hours first infusion, followed by 250 mg/m² IV over 60 minutes weekly[15] or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks[16] (preferred for every 2 weeks)

FOLFIRI + Panitumumab[17](*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes, Day 1
Repeat every 2 weeks

FOLFIRINOX[18,g]
Oxaliplatin 85 mg/m² IV Day 1,[s] Leucovorin 400 mg/m² IV over 2 hours on Day 1, Irinotecan 165–180 mg/m² IV over 30–90 minutes on Day 1, Fluorouracil 400 mg/m² IV Push Day 1, Fluorouracil 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46 hours) continuous infusion.
Repeat every 2 weeks

Modified FOLFIRINOX[19,20,g]
Oxaliplatin 85 mg/m² IV on Day 1,[s] Leucovorin 400 mg/m² IV over 2 hours on Day 1, Irinotecan 150 mg/m² IV over 30–90 minutes on Day 1, Fluorouracil 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46 hours) continuous infusion. Repeat every 2 weeks

FOLFIRINOX or Modified FOLFIRINOX + Bevacizumab[21,u]
Bevacizumab 5 mg/kg IV Day 1
Repeat every 2 weeks

IROX[22]
Oxaliplatin 85 mg/m² IV,[s]
followed by Irinotecan 200 mg/m² over 30–90 minutes every 3 weeks

IROX + Bevacizumab[u]
Bevacizumab 7.5 mg/kg IV Day 1
Repeat every 3 weeks

Bolus or infusional Fluorouracil/Leucovorin
Roswell Park regimen[23]
Leucovorin 500 mg/m² IV over 2 hours, Days 1, 8, 15, 22, 29, and 36
Fluorouracil 500 mg/m² IV bolus 1 hour after start of Leucovorin, Days 1, 8, 15, 22, 29, and 36
Repeat every 8 weeks

References

[c] An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.
[g] FOLFIRINOX is recommended instead of FOLFOXIRI because FOLFOXIRI uses a high dose of fluorouracil (3,200 mg/m² over 48 hours). Patients in the United States have been shown to have greater toxicity with fluorouracil. The dose of fluorouracil (2,400 mg/m² over 46 hours) is a starting dose consistent with the dose recommended in FOLFOX or FOLFIRI and should be strongly considered for U.S. patients.
[s] Oxaliplatin may be given either over 2 hours, or may be infused over a shorter time at a rate of 1 mg/m²/min. Leucovorin infusion should match infusion time of oxaliplatin. Cercek A, Park V, Yaeger R, et al. Faster FOLFOX: oxaliplatin can be safely infused at a rate of 1 mg/m²/min. J Oncol Pract 2016;12:e548-553.
[u] Bevacizumab may be safely given at a rate of 0.5 mg/kg/min (5 mg/kg over 10 minutes and 7.5 mg/kg over 15 minutes).

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PRINCIPLES OF PALLIATIVE SYSTEMIC THERAPY – REGIMENS AND DOSING[c]

Simplified biweekly infusional Fluorouracil/Leucovorin (sLV5FU2)[12]
Leucovorin[t] 400 mg/m² IV over 2 hours on Day 1,
followed by Fluorouracil bolus 400 mg/m² followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) continuous infusion
Repeat every 2 weeks
Weekly
Leucovorin 20 mg/m² IV over 2 hours on Day 1, Fluorouracil 500 mg/m² IV bolus injection 1 hour after the start of Leucovorin. Repeat weekly[24]
or
Fluorouracil 2600 mg/m² by 24-hour infusion plus Leucovorin 500 mg/m²
Repeat every week[24]

Bolus or infusional Fluorouracil + Bevacizumab[u]
Bevacizumab 5 mg/kg IV on Day 1
Repeat every 2 weeks

Capecitabine[25,v]
Capecitabine 850–1250 mg/m² PO twice daily for 14 days
Repeat every 3 weeks

Capecitabine + Bevacizumab[26,u]
Bevacizumab 7.5 mg/kg IV Day 1
Repeat every 3 weeks

Irinotecan
Irinotecan 125 mg/m² IV over 30–90 minutes, Days 1 and 8
Repeat every 3 weeks[27,28]
or Irinotecan 180 mg/m² IV over 30–90 minutes, Day 1
Repeat every 2 weeks
or Irinotecan 300–350 mg/m² IV over 30–90 minutes, Day 1
Repeat every 3 weeks

Irinotecan + Bevacizumab[31,u]
Irinotecan 180 mg/m² IV Day 1
Bevacizumab 5 mg/kg IV Day 1
Repeat every 2 weeks
or
Irinotecan 300–350 mg/m² IV Day 1
Bevacizumab 7.5 mg/kg IV Day 1
Repeat every 3 weeks

Irinotecan + Cetuximab (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² first infusion, followed by 250 mg/m² IV weekly[29]
or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks[16]
(preferred for every 2 weeks)

Irinotecan + Panitumumab[17,30] (*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes every 2 weeks

Cetuximab (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² first infusion, followed by 250 mg/m² IV weekly[29]
or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks[16]
(preferred for every 2 weeks)

Panitumumab[32] (*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes every 2 weeks

Regorafenib
Regorafenib 160 mg PO daily on Days 1–21[33]
or
First cycle: Regorafenib 80 mg PO daily on Days 1–7, followed by 120 mg PO daily on Days 8–14, followed by 160 mg PO daily on Days 15–21[34]
Subsequent cycles: Regorafenib 160 mg PO daily on Days 1–21
Repeat every 28 days

References

[c] An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.
[t] Leucovorin 400 mg/m² is the equivalent of levoleucovorin 200 mg/m².
[u] Bevacizumab may be safely given at a rate of 0.5 mg/kg/min (5 mg/kg over 10 minutes and 7.5 mg/kg over 15 minutes).
[v] The majority of safety and efficacy data for this regimen have been developed in Europe, where a capecitabine starting dose of 1000 mg/m² twice daily for 14 days, repeated every 21 days, is standard. Evidence suggests that North American patients may experience greater toxicity with capecitabine (as well as with other fluoropyrimidines) than European patients, and may require a lower dose of capecitabine.

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PRINCIPLES OF PALLIATIVE SYSTEMIC THERAPY – REGIMENS AND DOSING[c]

Trifluridine and Tipiracil ± Bevacizumab[35,36]
Trifluridine and Tipiracil 35 mg/m^2 up to a maximum dose of 80 mg per dose (based on the Trifluridine component)
PO twice daily Days 1–5 and 8–12
Bevacizumab 5 mg/kg on Days 1 and 15
Repeat every 28 days

Pembrolizumab[37] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
Pembrolizumab 2 mg/kg IV every 3 weeks
or Pembrolizumab 200 mg IV every 3 weeks
or Pembrolizumab 400 mg IV every 6 weeks
or Pembrolizumab 4 mg/kg IV every 6 weeks

Nivolumab[38] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
Nivolumab 3 mg/kg every 2 weeks
or Nivolumab 240 mg IV every 2 weeks
or Nivolumab 480 mg IV every 4 weeks
or Nivolumab 6 mg/kg IV every 4 weeks

Ipilimumab[39] + Nivolumab (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
Nivolumab 3 mg/kg (30-minute IV infusion) and Ipilimumab 1 mg/kg (30-minute IV infusion) once every 3 weeks for 4 doses, followed by Nivolumab 3 mg/kg IV or Nivolumab 240 mg IV every 2 weeks or Nivolumab 480 mg IV every 4 weeks

Dostarlimab-gxly[40] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
Dostarlimab-gxly 500 mg IV every 3 weeks for 4 doses followed by 1000 mg IV every 6 weeks

Cemiplimab-rwlc[41,42] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
350 mg IV on Day 1
Repeat every 3 weeks

Retifanlimab-dlwr[43,44] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
500 mg IV on Day 1
Repeat every 4 weeks

Tislelizumab-jsgr[45-48] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
200 mg IV on Day 1
Repeat every 3 weeks

Toripalimab-tpzi[49] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
3 mg/kg IV on Day 1
Repeat every 2 weeks

Penpulimab-kcqx[50] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
200 mg IV on Day 1
Repeat every 2 weeks

Pertuzumab[51] + Trastuzumab
(HER2-positive and *RAS* and *BRAF* WT)
Trastuzumab 8 mg/kg IV loading dose on Day 1 of cycle 1, followed by 6 mg/kg IV every 21 days
Pertuzumab 840 mg IV loading dose on Day 1 of cycle 1, followed by 420 mg IV every 21 days

Tucatinib[52] + Trastuzumab
(HER2-positive and *RAS* and *BRAF* WT),
Trastuzumab 8 mg/kg IV loading dose on Day 1 of cycle 1, followed by 6 mg/kg IV every 21 days
Tucatinib 300 mg PO twice daily

Fam-trastuzumab deruxtecan-nxki[53] (HER2-positive, IHC 3+)
Fam-trastuzumab deruxtecan-nxki 5.4 mg/kg IV on Day 1
Repeat every 21 days

References

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PRINCIPLES OF PALLIATIVE SYSTEMIC THERAPY – REGIMENS AND DOSING[c]

Encorafenib + Cetuximab[54-56] (*BRAF* V600E mutation positive)
Encorafenib 300 mg PO daily
Cetuximab 400 mg/m² IV followed by 250 mg/m² IV weekly
or Cetuximab 500 mg/m² IV every 2 weeks

Encorafenib + Panitumumab[54-56](*BRAF* V600E mutation positive)
Encorafenib 300 mg PO daily
Panitumumab 6 mg/kg IV every 14 days

FOLFOX/Encorafenib + Cetuximab[57] (*BRAF* V600E mutation positive)
Encorafenib 300 mg PO daily
Cetuximab 500 mg/m² IV Day 1
Oxaliplatin 85 mg/m² IV Day 1
Leucovorin 400 mg/m² IV Day 1
Fluorouracil 400 mg/m² IV bolus on Day 1, followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) IV continuous infusion
Repeat every 2 weeks

FOLFOX/Encorafenib + Panitumumab[57] (*BRAF* V600E mutation positive)
Encorafenib 300 mg PO daily
Panitumumab 6mg/kg IV every Day 1
Oxaliplatin 85 mg/m² IV Day 1
Leucovorin 400 mg/m² IV Day 1
Fluorouracil 400 mg/m² IV bolus on Day 1, followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) IV continuous infusion
Repeat every 2 weeks

Larotrectinib[58] (*NTRK 1/2/3* gene fusion-positive)
100 mg PO twice daily

Entrectinib[59] (*NTRK 1/2/3* gene fusion-positive)
600 mg PO once daily

Repotrectinib[60] (*NTRK 1/2/3* gene fusion-positive)
160 mg PO daily for first 14 days,
Then increase to 160 mg PO twice daily

Selpercatinib[61] (*RET* gene fusion-positive)
Patients ≥50 kg: 160 mg PO twice daily
Patients <50 kg: 120 mg PO twice daily

Adagrasib + Cetuximab[62] (*KRAS* G12C mutation positive)
Adagrasib 600 mg PO BID
Cetuximab 500 mg/m² IV every 2 weeks

Adagrasib + Panitumumab (*KRAS* G12C mutation positive)
Adagrasib 600 mg PO BID
Panitumumab 6 mg/kg IV every 2 weeks

Sotorasib + Cetuximab (*KRAS* G12C mutation positive)
Sotorasib 960 mg PO daily
Cetuximab 500 mg/m² IV every 2 weeks

Sotorasib + Panitumumab[63] (*KRAS* G12C mutation positive)
Sotorasib 960 mg PO daily
Panitumumab 6 mg/kg IV every 2 weeks

Fruquintinib[64]
5 mg PO daily on Days 1–21
Repeat every 28 days

References

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PRINCIPLES OF SURVIVORSHIP

Surveillance

- Surveillance recommendations can be found on APP-6 and APP-8.
- Long-term surveillance should be carefully managed with routine good medical care and monitoring, including cancer screening, routine health care, and preventive care.
- Routine CEA monitoring and routine CT scanning are not recommended beyond 5 years for high-risk disease. For low-risk disease, routine CEA monitoring and routine CT scanning are not recommended beyond 10 years.

Survivorship Care Planning

The oncologist and primary care provider should have defined roles in the surveillance period, with roles communicated to the patient.[1]

- Develop survivorship care plan that includes:
  - Overall summary of treatment, including all surgeries, radiation treatments, and systemic therapy received.
  - Description of possible expected time to resolution of acute toxicities, long-term effects of treatment, and possible late sequelae of treatment.
  - Surveillance recommendations.
  - Delineation of appropriate timing of transfer of care with specific responsibilities identified for primary care physician and oncologist.
  - Health behavior recommendations.
  - Fertility counseling.

Management of Late/Long-Term Sequelae of Disease or Treatment[2-6]

- For issues related to distress, pain, neuropathy, fatigue, or sexual dysfunction, see NCCN Guidelines for Survivorship.
- For chronic diarrhea or incontinence:
  - Consider anti-diarrheal agents, bulk-forming agents, diet manipulation, pelvic floor rehabilitation, and protective undergarments.
- Management of an ostomy:
  - Consider participation in an ostomy support group or coordination of care with a health care provider specializing in ostomy care (ie, ostomy nurse).
  - Screen for distress around body changes (NCCN Guidelines for Distress Management) and precautions around involvement with physical activity (see page SPA-C in the NCCN Guidelines for Survivorship).
- For oxaliplatin-induced neuropathy:
  - Consider duloxetine for painful neuropathy only, not effective for numbness, tingling, or cold sensitivity.[7]
  - Consider non-pharmacologic therapies such as heat or acupuncture.
  - Pregabalin or gabapentin are not recommended.

Counseling Regarding Healthy Lifestyle and Wellness[8]
(NCCN Guidelines for Survivorship)

- Undergo all age- and gender-appropriate cancer and preventive health screenings as per national guidelines.
- Maintain a healthy body weight throughout life.
- Adopt a physically active lifestyle (at least 30 minutes of moderate-intensity activity on most days of the week). Activity recommendations may require modification based on treatment sequelae (ie, ostomy, neuropathy).
- Consume a healthy diet with emphasis on plant sources. Diet recommendations may be modified based on severity of bowel dysfunction.
- Drink alcohol sparingly, if at all.
- Receive smoking cessation counseling as appropriate (NCCN Guidelines for Smoking Cessation).

Additional health monitoring and immunizations should be performed as indicated under the care of a primary care physician. Survivors are encouraged to maintain a therapeutic relationship with a primary care physician throughout their lifetime.

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American Joint Committee on Cancer (AJCC) TNM Staging Classification for Appendix Cancer 9th ed., 2022

Table 1. **Definitions for T, N, M**

T	**Primary Tumor**
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma *in situ* (intramucosal carcinoma; invasion of the lamina propria or extension into but not through the muscularis mucosae)
Tis(LAMN)	Low-grade appendiceal mucinous neoplasm confined to the muscularis propria; Acellular mucin or mucinous epithelium may invade into the muscularis propria T1 and T2 are not applicable to LAMN; Acellular mucin or mucinous epithelium that extends into the subserosa or serosa should be classified as T3 or T4a, respectively
T1	Tumor invades the submucosa (through the muscularis mucosa but not into the muscularis propria)
T2	Tumor invades the muscularis propria
T3	Tumor invades through the muscularis propria into the subserosa or the mesoappendix
T4	Tumor invades the visceral peritoneum, including acellular mucin or mucinous epithelium involving the serosa of the appendix or mesoappendix, and/or directly invades adjacent organs or structures
T4a	Tumor invades the visceral peritoneum, including acellular mucin or mucinous epithelium involving the serosa of the appendix or serosa of the mesoappendix
T4b	Tumor directly invades or adheres to adjacent organs or structures

N	**Regional Lymph Nodes**
NX	Regional lymph nodes cannot be assessed
N0	No tumor involvement of regional lymph node(s)
N1	Tumor involvement of one to three regional lymph nodes (tumor in lymph node measuring greater than or equal to 0.2 mm) or any number of tumor deposits is present with no tumor involvement in all identifiable lymph nodes
N1a	Tumor involvement of one regional lymph node
N1b	Tumor involvement of two or three regional lymph nodes
N1c	No tumor involvement of regional lymph nodes, but there are tumor deposits in the subserosa or mesentery
N2	Tumor involvement of four or more regional lymph nodes

M	**Distant Metastasis**
cM0	No distant metastasis
cM1	Distant metastasis
cM1c	Metastasis to sites other than peritoneum
pM1	Microscopic confirmation of distant metastasis
pM1a	Intraperitoneal acellular mucin, without identifiable tumor cells in the disseminated peritoneal mucinous deposits
pM1b	Intraperitoneal metastasis only, including peritoneal mucinous deposits containing tumor cells
pM1c	Microscopic confirmation of metastasis to sites other than peritoneum

Note: For specimens containing acellular mucin without identifiable tumor cells, efforts should be made to obtain additional tissue for thorough histologic examination to evaluate for cellularity.

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American Joint Committee on Cancer (AJCC)
TNM Staging System for Appendix Cancer 9th ed., 2022

Table 2. **Prognostic Groups**

	T	N	M
Stage 0	Tis	N0	M0
Stage 0	Tis(LAMN)		
Stage I	T1, T2	N0	M0
Stage IIA	T3	N0	M0
Stage IIB	T4a	N0	M0
Stage IIC	T4b	N0	M0
Stage IIIA	T1-T2	N1	M0
Stage IIIB	T3-T4	N1	M0
Stage IIIC	Any T	N2	M0
Stage IVA	Any T	Any N	M1a-M1b
Stage IVB	Any T	Any N	M1b
Stage IVC	Any T	Any N	M1c

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ABBREVIATIONS

A/P	abdomen/pelvis
AA	appendiceal adenocarcinoma
C/A/P	chest/abdomen/pelvis
CBC	complete blood count
CC	completeness of cytoreduction
CEA	carcinoembryonic antigen
CMP	comprehensive metabolic panel
CRS	cytokine release syndrome
ctDNA	circulating tumor DNA
dMMR	mismatch repair deficient
FDG	fluorodeoxyglucose
GBCA	gadolinium-based contrast agent
GCA	goblet-cell adenocarcinoma
HAMN	high-grade appendiceal mucinous neoplasm
IHC	immunohistochemistry
IPCT	intraperitoneal chemotherapy
LAMN	low-grade appendiceal mucinous neoplasm
MGPT	multigene panel testing
MiNEN	mixed neuroendocrine-nonneuroendocrine neoplasm
MMR	mismatch repair
MSI	microsatellite instability
MSI-H	microsatellite instability-high
MSS	microsatellite stable
MWA	microwave ablation
mut/Mb	mutations/megabase
NET	neuroendocrine tumor
NOS	not otherwise specified
PCI	Peritoneal Cancer Index
PNI	perineural invasion
RFA	radiofrequency ablation
SBRT	stereotactic body radiation therapy
TMB	tumor mutational burden
UC	undifferentiated carcinoma
WT	wild-type

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

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DISCUSSION UNDER DEVELOPMENT

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